BONE BUDS REV B | CONTROLLER | BARE IC ENGINEERING PROTOTYPE

RF matching, impedance, USB signal integrity, acoustic performance and cell safety require engineering qualification.

Controller requires protected 1S 4.2V LiPo pack + 10k NTC. USB defaults 100mA; firmware may select 500mA ONLY after enumeration.

Audio: 64 clocks per stereo frame. Separate microphone input and amplifier output I2S buses. BTL speaker terminals must never be grounded.

U1
ESP32-S3R8

U2
W25Q128JVFIQ / 16 MB

U3
USBLC6-2SC6

U4
BQ24074RGT

U5
TPS22917DBV

U6
TPS63531DSK

U7
TPS7B3933DBZR / 3.08V

R1
15R

C1
1u

C2
1u

Y1
40MHz 10ppm CL8pF

L1
2nH RF tuning

C3
12p COG tune

C4
12p COG tune

R2
10k

C5
1u

R3
10k

R4
100k

TP1
BOOT

TP2
CHIP_EN

TP3
GND

TP4
UART_TX

TP5
UART_RX

C6
100n

C7
100n

C8
100n

C9
100n

C10
100n

C11
10u

C12
22u

FB1
BLM18AG601SN1 600R

C13
1.5p RF tune

L2
2.4nH RF tune

C14
1.5p RF tune

J1
U.FL 2.4GHz antenna

J2
GCT USB4105-GF-A

R5
5.1k 1%

R6
5.1k 1%

R7
22R

R8
22R

R9
3.57k 1% 2x99mA

R10
3.48k 1%

C15
4.7u 10V

C16
4.7u 10V

C17
10u 10V

J3
JST SH 3P

TP6
BAT

TP7
5V5

C18
1n

C19
10u

R11
100k

SW1
POWER

L3
1.0uH >=2A DCR<100m

C20
10u

C21
100n

C22
10u

C23
10u

D1
BAT54H

R12
1M

R13
330k

C24
100n

R14
33R

R15
33R

R16
33R

R17
33R

R18
33R

R19
100k

R20
100k

SW2
MUTE / PLAY

J4
JST SH 10P

J5
JST SH 10P

J6
JST SH 2P

R21
10k

TP8
3V3

TP9
CHG

TP10
PGOOD

R22
10k

R23
10k

R24
100R

#FLG79
PWR_FLAG

#FLG80
PWR_FLAG

#FLG81
PWR_FLAG

#FLG82
PWR_FLAG

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File: controller.kicad_sch

Title: Bone Buds Rev B — controller

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